Curated to Win: Certifying Reward-Model and Judge Selection Against Evaluation-Set Curation

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Picking an evaluator

- Research pipelines need an evaluator: a reward model or an LLM judge
- They pick one by its accuracy on a validation set
- But the pipeline itself often decides which items go in that set

How much of the validation set must be removed before a different evaluator wins?

Example: RewardBench

All 2,985 items

Model	Score
INF-ORM-Llama3.1-70B	95.11
QRM-Gemma-2-27B	94.44
Skywork-Reward-Gemma-2-27B-v0.2	94.26

After removing 12 items

Model	Score
QRM-Gemma-2-27B	95.11
INF-ORM-Llama3.1-70B	95.05
Skywork-Reward-Gemma-2-27B-v0.2	94.80

Removing 12 of 2,985 items puts the runner-up first.

118 models. Scores use RewardBench’s own scoring code.

Setup

Official score of model *m*

$$S_m = \sum_i w_i^0 \, r_{mi}$$

A **curation** keeps a subset *K* of the items and rescores every model on it

$$S_m^K = \frac{1}{f} \sum_{i \in K} w_i^0 \, r_{mi}, \qquad f = \sum_{i \in K} w_i^0$$

K **crowns** *m* if *m* is strictly first on *K*

Gap from the winner *a* to a challenger *c*

$$g_c = S_a - S_c$$

Certificate in weight

Any curation that lets *c* tie or beat the winner removes a share of at least its gap:

$$S_c^K \geq S_a^K \implies 1 - f \geq g_c$$

Each removed item closes the gap by at most its own weight:

$$f\,(S_a^K - S_c^K) = g_c - \sum_{i \notin K} d_i^c \;\geq\; g_c - (1 - f)$$

So the winner is safe against any removal below $g^\star = \min_c g_c$

This is the gap to the runner-up, which you can read off a leaderboard.

Certificate in items

Let each item's contribution to the winner's lead be $d_i^c = w_i^0 \left(r_{ai} - r_{ci}\right)$

c overtakes *a* exactly when the removed contributions sum past the gap:

$$S_c^K > S_a^K \iff \sum_{i \notin K} d_i^c > g_c$$

So sort the contributions from largest to smallest and count:

$$k_c = \min\left\{k : \sum_{j \le k} d_{(j)}^c > g_c\right\}, \qquad k^\star = \min_c \; k_c$$

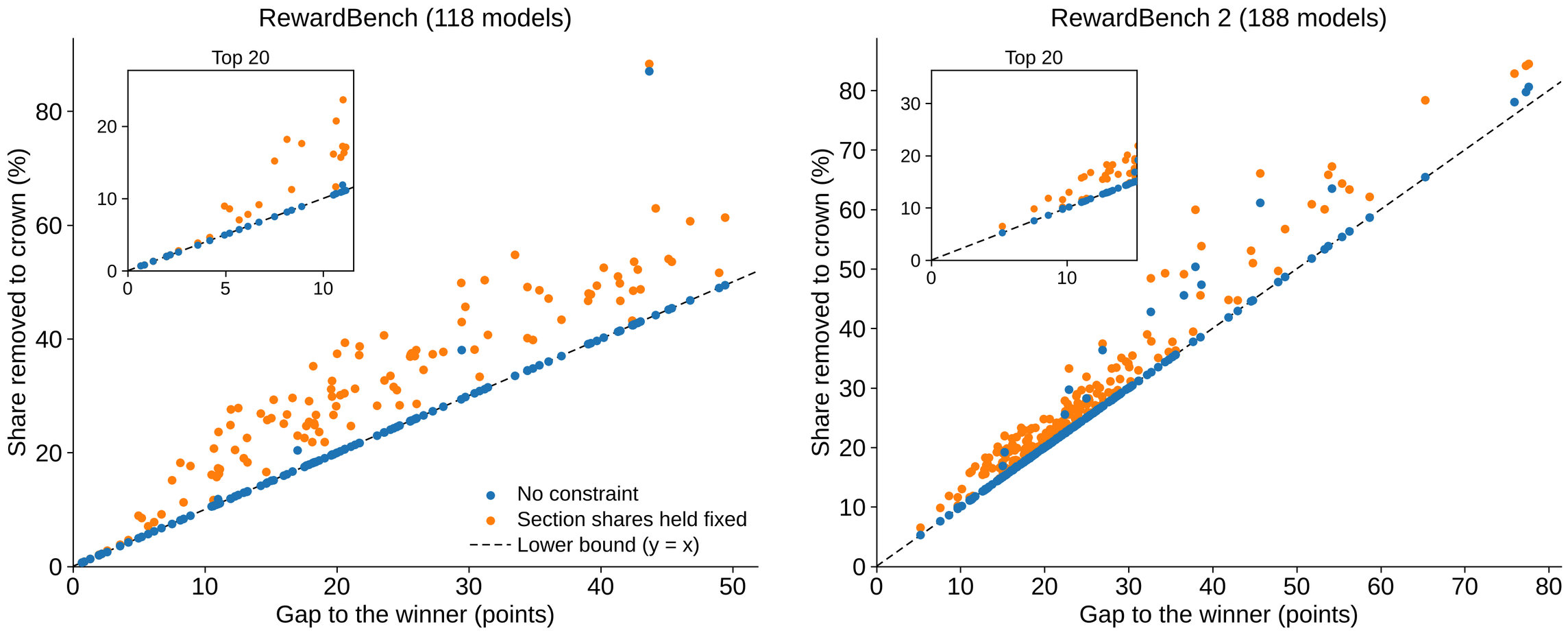
This is exact: any *k** − 1 items are safe, and some *k** items change the winner.

Results

	RewardBench	RewardBench 2
Models	118	188
Items	2,985	1,753
Top-two gap (points)	0.67	5.22
Cheapest curation found (% of weight)	0.68	5.30
Items that change the winner (k*)	12	43
Same, under its own scoring code	11	40

RewardBench 2 scores exclude the Ties subset. Curation costs are upper bounds.

How tight is the weight bound?



For the top 20 RewardBench models, the cheapest curation found costs at most 1.02 times the gap.

Possible winners

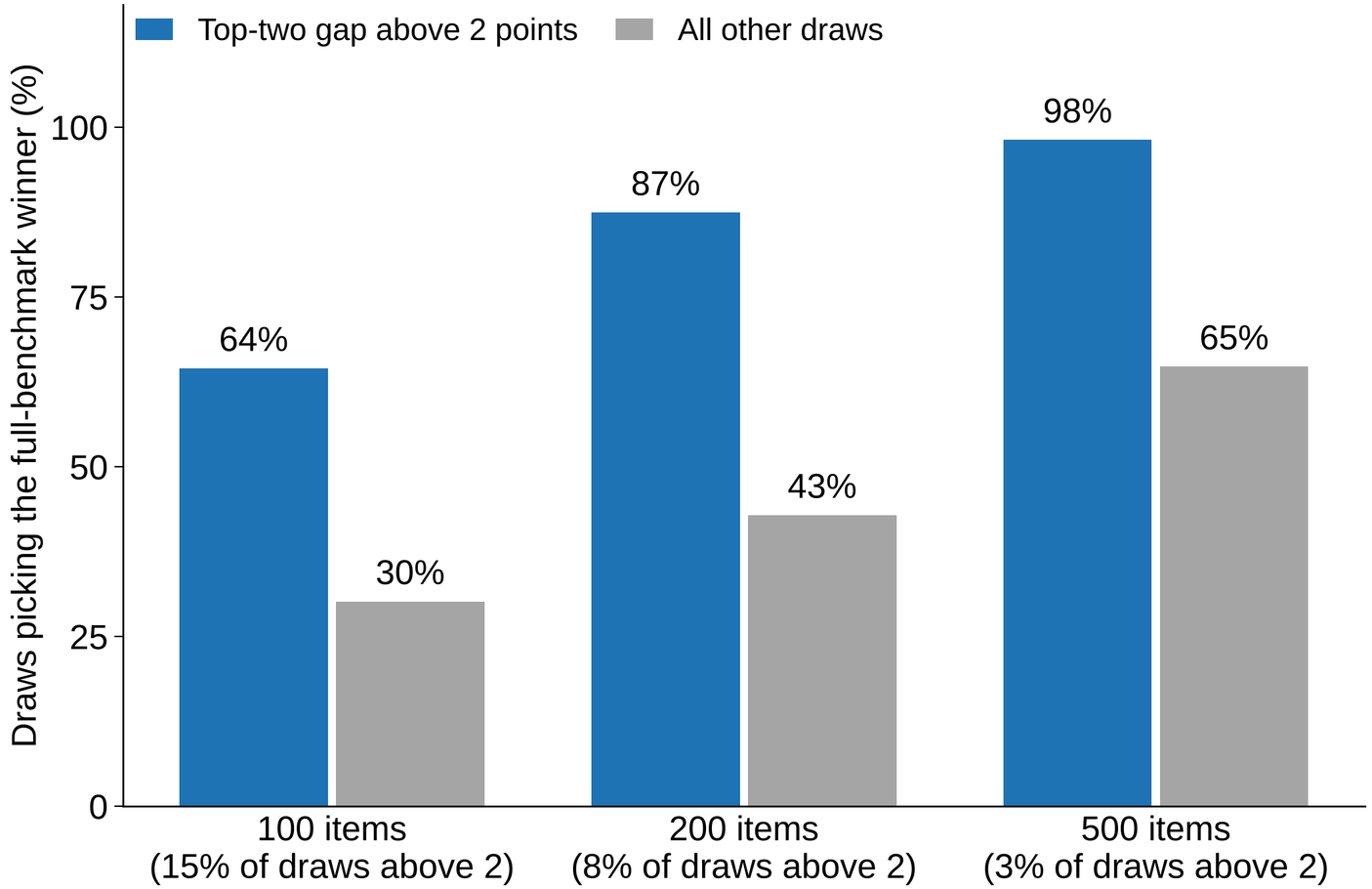
If up to a share δ of the benchmark may have been curated, how many models could be the winner?

Budget δ	0.5%	1%	2%	5%	10%
RewardBench	1	3	5–6	11	19
RewardBench 2	1	1	1	1	6

At a 1% budget, a pipeline can only say the best RewardBench model is one of 3.

5–6 means 5 models have a verified curation and 6 are within δ of the winner.

Small validation sets



- On a small set, the certificate is its top-two gap
- Random RewardBench sets, 2,000 draws per size
- This measures sampling noise, not curation

A protocol for selection

1. Fix the items and record a hash before scoring anything. Score every candidate on all of them, and log every later filter.
2. Set a curation budget δ.
3. Report g^* and k^*. Accept the winner only if $g^* > \delta$.
4. Otherwise, report the set of possible winners.

RewardBench at a 1% budget: $g^* = 0.67$ points, so report 3 possible winners.

Summary

- Two certificates for a selected evaluator: g^* in weight and k^* in items
- On RewardBench, 12 of 2,985 items change the winner
- For top models, the top-two gap is close to the true cost

Limitations

- Removal only, not reweighting or paraphrasing
- Weighted accuracy only, not mean win rate
- LP costs are upper bounds; the item count is exact
- I do not claim any published result was curated

github.com/adrian-erlikhman/curated-to-win

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